CS 2840 Fall 2025 Scribes: Pranav Ajit Nair, Yoel Zimmermann
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Lecture 03: Kantorovich, Couplings

1 Introduction

Today we will talk about another formulation of OT by Kantorovich, which introduces *couplings* as a central concept.

1.1 Recap

In the last class, we introduced the concept of measures, push-forwards, the Monge formulation in both the discrete (boils down to a matching probelm) and the continuous (mapping) versions. We also talked about Brenier's theorem, which guarantees existence and uniqueness of the solution to Monge under simple assumptions. In addition, we discussed the failure modes of the Monge formulation: i) there is no way to split mass, ii) there is a notion of asymmetry, and iii) the solution is generally not unique, i.e., there may be several solutions with the same cost.

Today we will see Brenier's theorem again, but for Kantorovich's formulation.

2 Kantorovich Relaxation

Leonid Kantorovich: Leonid Kantorovich was a mathematician and economist and one of the main founders of linear programming and optimal resource allocation. In 1942, he introduced the relaxation of the Monge problem. For this and related work, he later shared the 1975 Nobel Prize in Economics with Tjalling C. Koopmans. Fun fact: the CIA kept a dossier on Kantorovich. Kantorovich's contributions reflected a broader trend, as many advances in optimization were driven by the demands of WW2.

Consider the previous example where coal mines with supply $a_1, \dots, a_n$ are sending coal to factories with demand $b_1, \dots, b_n$. In Monge's formulation, each mine must send all its coal to exactly one factory (one-to-one mapping). This is restrictive. Kantorovich's idea was to allow splitting: mine i can send a fraction of its coal π_{ij} to factory j. We call the matrix π_{ij} a coupling between $\mathbf{a}$ and $\mathbf{b}$.

Couplings formalize the notion of splitting mass.

Definition 1. *Suppose we have measures*

$$\alpha = \sum_{i=1}^{n} a_i \delta_{x_i}, \quad a \in \Delta_n$$

$$\beta = \sum_{j=1}^{m} b_j \delta_{y_j}, \quad b \in \Delta_m,$$

where Δ_n and Δ_m are probability simplexes. The coupling $\pi \in \mathbb{R}_+^{n \times n}$ is a valid coupling if it has marginals α and β, i.e., the rows sum to $\mathbf{a}$ and the columns sum to $\mathbf{b}$.

We denote the set of all valid couplings by $\Pi(\mathbf{a}, \mathbf{b}) = \left\{\pi \in \mathbb{R}_+^{n \times m} \mid \pi \mathbf{1}_m = \mathbf{a}, \pi^\top \mathbf{1}_n = \mathbf{b}\right\}$. This set is called the transportation polytope. Intuitively, it represents all possible transport plans that exactly redistribute the mass of α into β. Notice that π is also a joint probability distribution on $X \times Y$, since $\sum_i \sum_j \pi_{ij} = 1$.

Example 1. *Permutations. $\sigma \in S_n \mapsto [\pi_\sigma]_{ij}$. For example, $\sigma(1) = 2, \sigma(2) = 3, \sigma(3) = 1 \mapsto 1/3 \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.*

Example 2. *Uniform (independent) coupling.* $\pi_{ij}^{\text{unif}} = a_i b_j$, *i.e.,* $\pi^{unif} = \mathbf{a}\mathbf{b}^\top$

Having defined couplings, we can now state Kantorovich's formulation. The key idea is: among all admissible couplings, choose the one that minimizes the total transport cost. If $(C_{i,j})_{i,j=1}^{n,m} \in \mathbb{R}^{n\times m}$ is the cost matrix and given the set of admissible couplings $\Pi(\mathbf{a},\mathbf{b}) = \left\{\pi \in \mathbb{R}_+^{n\times m} \mid \pi \mathbf{1}_m = \mathbf{a}, \pi^\top \mathbf{1}_n = \mathbf{b}\right\}$, the Kantorovich problem is:

$$\min_{\pi\in\Pi(\mathbf{a},\mathbf{b})} \sum_{i=1}^{n}\sum_{j=1}^{m} C_{i,j}\pi_{i,j}. \tag{1}$$

This is a linear program: the objective is linear in π and the constraints are linear. There are several equivalent ways to write it:

(1) Matrix form:
$$\begin{aligned} \min_{\pi\in\mathbb{R}^{n\times m}} \quad & \sum_{i=1}^{n}\sum_{j=1}^{m} C_{ij}\pi_{ij} \\ \text{s.t.} \quad & \sum_{j=1}^{m} \pi_{ij} = a_i, \quad \forall i = 1,\dots,n, \\ & \sum_{i=1}^{n} \pi_{ij} = b_j, \quad \forall j = 1,\dots,m, \\ & \pi_{ij} \geq 0, \quad \forall i,j. \end{aligned}$$

(2) Vectorized form:
$$\begin{aligned} \min_{x\in\mathbb{R}^{nm}} \quad & c^T x \\ \text{s.t.} \quad & (I_n \otimes 1_m^T)\, x = a, \\ & (1_n^T \otimes I_m)\, x = b, \\ & x \geq 0. \end{aligned}$$

(3) Inner product form:
$$\begin{aligned} \min_{\pi\in\mathbb{R}^{n\times m}} \quad & \langle C, \pi\rangle = \operatorname{tr}(C^T\pi) \\ \text{s.t.} \quad & \pi 1_m = a, \\ & \pi^T 1_n = b, \\ & \pi \geq 0. \end{aligned}$$

Particular case: Uniform measure. Assume $n = m$, $\alpha = \sum_1^n \frac{1}{n}\delta_{x_i}, \beta = \sum_1^n \frac{1}{n}\delta_{y_j}$. The valid couplings in this case are the bistochastic (or doubly-stochastic) matrices $\sum_i \pi_{ij} = \sum_j \pi_{ij} = \frac{1}{n}$. The transporation polytope is then called the Birkhoff polytope, which is denoted $\mathscr{B}_n$. This is relevant in most ML use-cases.

Also, note that $\text{Perm}(n) \subset \Pi(\mathbf{a},\mathbf{b})$. This implies that Kantorovich solution will always be better or equally good as a solution to the Monge problem (the bigger the feasible set, the better we can optimize).

Theorem 1. ***Birkoff-von Neumann Theorem****. Every* $M \in \mathscr{B}_n$ *can be written as a convex combination of permutation matrices:* $M = \sum_{k=1}^{K} \lambda_k P_k, \quad \lambda_k \geq 0, \quad \sum_k \lambda_k = 1.$

The proof can be found on p.15 in [**?**]. The theorem implies that the Kantorovich formulation can be seen as a convex relaxation of Monge's problem: the feasible set is enlarged from permutation matrices to their convex hull (the Birkhoff polytope). Intuitively, a coupling can be understood as an interpolation between permutations. This guarantees that a solution always exists, and the optimization becomes convex.

3 Comparing Monge with Kantorovich.

The main differences between the two formulations are summarized in Table 1.

Aspect	Monge	Kantorovich
Variables	n	n^2
Constraints	Permutation Constraints	Marginals Fixed
Solution	Permutation (matching)	Coupling (transport plan)
Existence	Not always solvable.	Always solvable (feasible set bounded, objective linear).
Uniqueness	Can have multiple optimal solutions (non-LP structure).	Can have multiple optimal solutions (as with LPs).
Feasible region	Non-convex.	Convex (convex hull).
Symmetry	Asymmetric: T pushforward $a \to b$ does not imply T^{-1} is valid.	Symmetric: $\pi \in \Pi(\mathbf{a}, \mathbf{b}) \iff \pi^\top \in \Pi(\mathbf{b}, \mathbf{a})$.
Complexity	Assignment problem over n variables: $O(n^3)$.	LP over $n \times n$ couplings: $O(n^6)$ (naive), $O(n^3)$–$O(n^4)$ with specialized solvers. Typically slower in practice.
Takeaway	Simple, fast, but not always well-posed.	Flexible, always well-posed, but more complex.

Table 1: Comparison of Monge and Kantorovich Optimal Transport Formulations

4 Continuous Couplings

So far we have worked in the discrete setting. To move to the continuous case, we replace the coupling matrix π with a joint probability measure $\pi \in \mathscr{P}(X \times Y)$. Each value $\pi(x, y)$ is an infinitesimal density over pairs (x, y). The constraints on row and column sums become the constraints

$$P_{\mathscr{X}\sharp}\pi = \alpha$$
$$P_{\mathscr{Y}\sharp}\pi = \beta$$

where $P_{\mathscr{X}\sharp}$ is pushforward of projection $P_{\mathscr{X}}(x, y) = x$, which we can equivalently write as

$$\pi(A \times \mathscr{Y}) = \alpha(A), \pi(\mathscr{X} \times B) = \beta(B).$$

The objective now is:

$$\min_{\pi \in \Pi(\alpha,\beta)} \int_{X \times Y} c(x, y)\, d\pi(x, y)$$
$$\Pi(\alpha, \beta) = \{\pi \in P(X \times Y) \mid P_{X\#}\pi = \alpha, P_{Y\#}\pi = \beta\}$$

and can be interpreted as the average work done to move the mass.

More generally, couplings can appear in three flavors depending on whether the source and target distributions are discrete or continuous. When both are discrete, the coupling is a matrix $\pi \in \mathbb{R}_+^{n \times m}$. When both are continuous, the coupling is a measure $\pi \in \mathscr{P}(X \times Y)$. In the mixed discrete–continuous case, each discrete point corresponds to a probability distribution over the continuous space, which one can think of as a vector of densities.

5 Revisiting Brenier's Theorem

Now, we revisit Brenier's theorem in the context of the Kantorovich relaxation. We have the same assumptions as before, but we obtain a stronger result.

Theorem 2. *Suppose $c(x, y) = \|x - y\|^2$ and α has a density $\rho_\alpha = \frac{d\alpha}{dx}$. Then, there exists a unique coupling π that solves the Kantorovich problem. This coupling is deterministic, supported on the graph of a Monge map $\pi = (\mathrm{Id}, T)_\sharp\alpha$. T is the gradient of a convex function.*

A sketch of the proof can be found on p.27-28 in [**?**]. This formulation also provides conditions under which Monge and Kantorovich problems are equivalent. In such cases, Monge maps have a very special structure, which is exploited by recent neural OT methods.

6 ML Applications of Kantorovich

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- **Domain adaptation: matching datasets.** Align source and target domains when marginal distributions differ. Example: reweight source training samples to match target distribution. Coupling tells us **?**which source points map to which target points**?** (probabilistically).
- **Embedding alignment.** Match vector spaces across languages or modalities. Example: bilingual word embeddings (align English & Italian spaces without parallel data). Couplings capture soft correspondences between vocabularies.
- **Uncertainty-aware mappings via probabilistic transport.** Unlike Monge maps, couplings allow splitting mass → captures uncertainty. Useful when multiple plausible correspondences exist (e.g., many-to-many) Applications: shape matching under noise, multimodal prediction.

References